

Improved Performance of GaN/AlN Resonant Tunneling Diodes with Current Densities
exceeding 1 MA/cm²

Thesis

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By

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Abstract

Identical GaN/AlN resonant tunneling diode (RTD) structures were grown on free-standing bulk GaN at substrate temperatures of 760 °C, 810 °C, 860 °C, and 900 °C via plasma-assisted molecular-beam-epitaxy (MBE). Each sample displayed negative differential resistance (NDR) at room temperature. The figures-of-merit quantified were peak-to-valley-current-ratio (PVCR), yield of device with room-temperature NDR, and peak current density (J_p). The figures-of-merit demonstrate an inverse relationship between PVCR/yield and J_p over this growth temperature series. X-ray diffraction and transmission electron microscopy were used to determine the growth rates and layer thicknesses used to explain the varying figures-of-merit. Due to the high yield of devices grown at 760 °C and 810 °C: PVCR, peak voltage (V_p), and J_p were plotted versus device area and demonstrated high uniformity and application tunability. Peak current densities of up to 1.01 MA/cm² were observed for the sample grown at 900 °C.

Dedication

This work is dedicated to my family and Krista

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1. Introduction

Over the past decade, there has been a considerable effort to produce high-performance GaN-based resonant tunneling diodes (RTD), fueled by the strong demand for affordable, compact, high power millimeter-wave (mm-wave) and terahertz (THz) devices. Applications for such devices are found in spectroscopy, imaging, security, medicine, high-resolution sensing, and broadband communications. RTDs are promising candidates to fill these needs since they are based on ultra-fast electron transport by means of quantum mechanical tunneling. Fundamental oscillations of 712 GHz ^[1] and 1.92 THz ^[2] at room temperature have already been reported in InAs/AlSb and InGaAs/AlAs RTD devices, but these lacked significant output power. Recently, using on-chip biasing and power combining, 1 mW of output power at 260 GHz from an InGaAs-based RTD has been achieved. ^[3] III-nitride-based RTDs have an inherent potential to achieve higher output power due to their considerably wider band-gap energies. Unfortunately, while great progress has been made ^[4-19], GaN-based RTDs are not yet well enough understood to produce devices capable of reaching oscillation frequencies above 1 GHz. ^[16]

The output power of an RTD-based oscillator can be estimated by $\frac{3}{16} \Delta I \Delta V$, where ΔI and ΔV are the differences between the peak and valley currents and voltages, respectively, of the NDR region. ^[21] Recently, there have been reports of peak tunneling current densities in GaN-based RTDs ranging from 100 kA/cm² up to 930 kA/cm².

[11,15,16,18,19] However, their respective peak-to-valley current ratios (PVCRs) are low, which translates to small ΔI . Another recent study has reported a PVCR above 2 in a GaN/AlN RTD, but the current density was relatively low at 30 kA/cm². [22] However, that paper and a previous paper, [23] reported 10%-90% switching rise times of 33 ps and 55 ps, respectively, which correspond to fundamental oscillation frequencies in the tens of GHz range. In order to reach frequencies in the THz range, GaN-based RTDs must be fundamentally better understood. To accomplish this, a growth temperature series of GaN/AlN-based RTDs was performed and devices were fabricated and characterized for electrical properties. The results for the study are presented in Chapter 2. Chapter 3 provides a preliminary look into Ga₂O₃/Al₂O₃ based RTDs as this emergent material system has the desirable qualities of a high breakdown field, wide bandgap and large Baliga's figure-of-merit for high-power applications. Before presenting the experimental results, background information, fundamental operation and RTD properties will be discussed.

1.1: Introduction to Tunneling

Tunneling is fundamentally quantum mechanical. It is a phenomenon that allows for particles to pass through an energy barrier with larger potential than the energy of the particle. This is a forbidden event from a classical physics point-of-view; a particle can only surmount an energy barrier by having been set in motion with greater energy than the barrier. Quantum mechanics states particles have wave-like nature and is described by a particle's wave function, $\psi(x)$, the modulus square of which gives the probability amplitude of finding the particle in space. Additionally, Heisenberg's uncertainty principle

postulates the position and momentum of a particle cannot be measured exactly and simultaneously. These two properties manifest into quantum mechanical tunneling, as a particle's wave function can span all space and give non-zero probabilities to measuring the particles position on the “other side” of a potential barrier.

1.2: Resonant Tunneling

Resonant tunneling can be explained by combining the ideas from two separate stationary potential-well problems, each solved quantum mechanically by means of the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + [V(x) - E]\psi(x) = 0 . \quad (1.1)$$

The first problem is to find the tunneling transmission probability of a free particle with energy less than a singular barrier of finite width and energy. The second problem is to find the bound energy states in a square potential well with walls of finite energy but extend spatially to infinity. Resonant tunneling is thus the combination of these two potential-well problems were the finite width barriers of the first replace the spatially infinite walls of the second. The two problems will be discussed first separately as they contain the root ideas of quantum mechanics application, followed by a discussion of the double barrier resonant tunneling problem.

1.2.1: Tunneling Through a Single Barrier

The first problem can be split into three regions: Region I is the space ($x < -a$), Region II is the space ($-a < x < a$), and Region III is the space $x > a$. A plot of the potential and tunneling wavefunctions is shown in Figure 1.1. The whole problem can be

solved as a coupled set of differential equations, one for each region, with the proper boundary conditions at $\pm a$. In quantum mechanics, Max Born postulated that the wave function for a particle or system must be: (i) Single valued over all space, (ii) square-integrable over all space, (iii) continuous over all space and (iv) finite valued second derivatives, or stated in another way, continuous first derivatives over all space. These four postulates provide the appropriate boundary conditions to completely solve this (and the following two) problems.

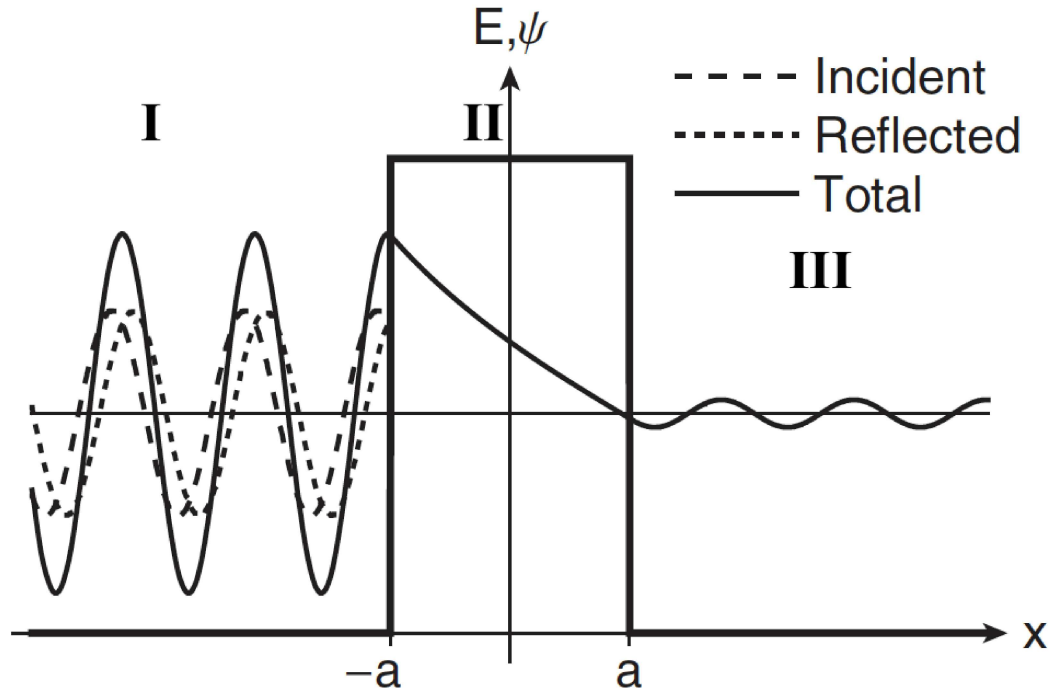


Figure 1.1 Single barrier potential with incident, reflected and transmitted wavefunctions. Adapted from *Quantum Mechanics: A paradigms approach* [30]

At first glance, the three regions can be organized into two cases. Regions I and III belong to Case I, where the particles energy (E) is greater than the potential energy (V) and Region II belongs to Case II, where the particles energy (E) is less than the potential energy (V). Case I involves free-particles and Case II involves particles that are tunneling. The

two cases are more general than the regions as they apply to all potential-well problems and so they will be first solved for generally, then applied specifically to the regions in this problem.

To begin, the time-independent Schrödinger equation will be solved generally for a constant stationary potential, V .

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m(V - E)}{\hbar^2} \psi(x) \quad (1.2)$$

Employing an Ansatz: $\psi = Ae^{ikx} + Be^{-ikx}$, Equation 1.2 reduces to the following:

$$-k^2\psi(x) = \frac{2m(V - E)}{\hbar^2} \psi(x) \quad (1.3)$$

$$k = \frac{\sqrt{2m(E - V)}}{\hbar}. \quad (1.4)$$

The solution for the wavefunction is therefore: $\psi = Ae^{ikx} + Be^{-ikx}$ where 'k' is expressed in Eq. 1.4. Now a general solution has been obtained and can be applied to the two cases discussed above.

Case I: The energy of the particle is greater than the potential it resides in. One thing to note is the ability to change the reference point for potential energies; only the difference in energy occurs in the equations. On this note, the potential energy can be set to zero with an equal reduction in the absolute value of the particle's energy. The solution for Case I is described as:

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \quad (1.5)$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

Case II: The energy of the particle is less than the potential it resides in. This produces an imaginary value of 'k' since $E - V$ is negative. To fully reduce the solution, i , is factored from the numerator of the expression. The solution for Case II is therefore:

$$\psi(x) = Ce^{-\kappa x} + De^{\kappa x} \quad (1.6)$$

$$\kappa = \frac{\sqrt{2m(V - E)}}{\hbar}$$

Now the problem can easily be solved by applying Eq. 1.5 and 1.6 along with the boundary conditions to the three regions. The general wavefunction for the problem and accompanying boundary condition equations are as follows:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{for } (x < -a) \\ Ce^{-\kappa x} + De^{\kappa x} & \text{for } (-a < x < a) \\ Fe^{ikx} + Ge^{-ikx} & \text{for } (x > a) \end{cases} \quad (1.7).$$

For the continuity of the wave function at $x = -a$ and $x = a$:

$$Ae^{-ika} + Be^{ika} = Ce^{\kappa a} + De^{-\kappa a} \quad (1.8)$$

$$Ae^{-ika} - Be^{ika} = \frac{i\kappa}{k} (Ce^{\kappa a} - De^{-\kappa a}) \quad (1.9)$$

$$Ce^{-\kappa a} + De^{\kappa a} = Fe^{ika} + Ge^{-ika} \quad (1.10)$$

$$De^{\kappa a} - Ce^{-\kappa a} = \frac{ik}{\kappa} (Fe^{ika} - Ge^{-ika}) \quad (1.11).$$

A simplification can be implemented by allowing only incident waves from Region I. The coefficients 'A' and 'F' correspond to the magnitude of waves moving in the positive x-direction, 'B' and 'G' correspond to the magnitude of waves moving in the negative x-

direction, and ‘C’ and ‘D’ correspond to the decreasing and increasing exponential functions in the barrier region. Thus, by only allowing incident waves from Region I, ‘B’ can be defined as the amplitude of the wave reflected off the barrier, ‘F’ the amplitude of the transmitted wave and ‘G’ can be set to zero. The other coefficients remain the same.

The transmission coefficient is defined as $T = \left| \frac{F}{A} \right|^2$ and the reflection coefficient is defined

as $R = 1 - T = \left| \frac{B}{A} \right|^2$. Solving the coupled set of equations 1.8-1.11 for these yields:

$$T = \frac{1}{1 + \frac{(k^2 + \kappa^2)^2}{4(k\kappa)^2} \sinh^2 2\kappa a} = \frac{1}{1 + \frac{V^2}{4E(V-E)} \sinh^2 \frac{2a}{\hbar} \sqrt{2m(V-E)}} \quad (1.12)$$

$$R = 1 - T = \frac{1}{1 + \frac{4(k\kappa)^2}{(k^2 + \kappa^2)^2} \sinh^2 2\kappa a} = \frac{1}{1 + \frac{4E(V-E)}{V^2 \sinh^2 \frac{2a}{\hbar} \sqrt{2m(V-E)}}} \quad (1.13)$$

The transmission probability is non-zero when the incident energy is less than the barrier height and describes the action of single barrier tunneling.

1.2.2: Quasibound states in finite potential well

The second problem is to find the allowed energy states within a finitely valued square potential well. This problem reveals the nature of a quantum well and is a fundamental property and requirement for the operation of a double barrier resonant tunneling diode. Problem two can be approached similarly to the first by creating three regions and applying the general wavefunction forms of Case I and II. Regions I and II are the sets $(x < -a)$ and $(x > -a)$, respectively, with potential energy V . Region II is the set $(a < x < -a)$ and is the potential zero. A plot of the potential is shown in Figure 1.2.

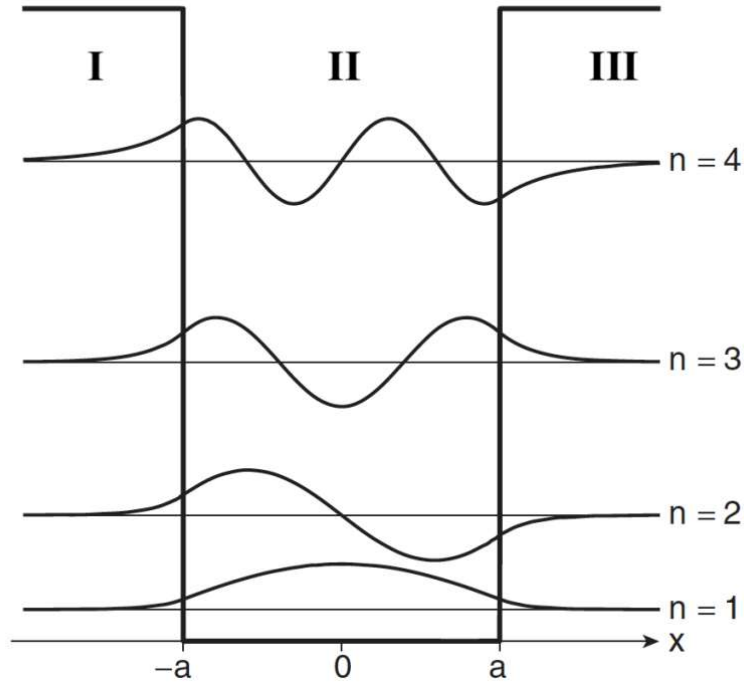


Figure 1.2 A finite square potential well with the first four quasi-bound eigen-energies. Adapted from *Quantum Mechanics: A paradigms approach* [30]

The general wavefunction form is:

$$\psi(x) = \begin{cases} Ae^{\kappa x} + Be^{-\kappa x} & x < -a \\ Ce^{ikx} + De^{-ikx} & -a < x < a \\ Fe^{\kappa x} + Ge^{-\kappa x} & x > a \end{cases} \quad (1.14).$$

Three simplifications can be made for this problem: (i) The potential well is symmetrical about $x = 0$ and allows the breaking up of the problem into even and odd function solutions; (ii) Region II does not require traveling waves and can be reduced to stationary functions; (iii) The Born postulate requiring the wave function be square-integrable requires the wave function to decrease at positive and negative infinity and is done by setting 'B' and 'F' to zero. The wavefunction is now:

$$\psi_{\text{even}}(x) = \begin{cases} Ae^{\kappa x} & x < -a \\ D \cos kx & -a < x < a \\ Ae^{-\kappa x} & x > a \end{cases} \quad (1.15)$$

$$\psi_{odd}(x) = \begin{cases} Ae^{\kappa x} & x < -a \\ C \sin kx & -a < x < a \\ -Ae^{-\kappa x} & x > a \end{cases} \quad (1.16)$$

Employing the boundary conditions at $x = a$ yields the following equations required to solve for the eigen-energies and coefficients:

$$D \cos ka = Ae^{-\kappa a} \quad (1.17)$$

$$-kD \cos ka = -\kappa Ae^{-\kappa a} \quad (1.18)$$

$$C \sin ka = -Ae^{-\kappa a} \quad (1.19)$$

$$kC \sin ka = \kappa Ae^{-\kappa a} \quad (1.20).$$

Equations for the even function energies, 1.17 and 1.18, and the equations for the odd function energies, 1.19 and 1.20, reduce to the following equations needed to solve for eigen-energies:

$$\sqrt{\frac{2mE}{\hbar}} \tan a \sqrt{\frac{2mE}{\hbar}} = \sqrt{\frac{2m(V-E)}{\hbar}} \quad (1.21)$$

$$-\sqrt{\frac{2mE}{\hbar}} \cot a \sqrt{\frac{2mE}{\hbar}} = \sqrt{\frac{2m(V-E)}{\hbar}} \quad (1.22).$$

The equations require numerical methods for eigen-energy solutions with specific values for the well width and potential energy. Never-the-less, these results show the existence of bound states in a system with physical analog.

1.2.3: Double finite-barrier Potential Well

The ideas of tunneling through barriers and the existence of quasi-bound states previously discussed lay the foundation for a double-barrier potential well problem and the base for a double-barrier resonant tunneling diode. A short look into the ideal double barrier

potential well problem will reveal a more desirable property than the separate notions of bounded or tunneling particles. In single barrier tunneling, the transmission probability only becomes large when the incident energy is close to the top of the potential barrier. The potential well problem showed the ability to trap particles in potential wells at quantized energy levels. From the point of view of a device, specifically the RTD, it is not desirable to trap particles or only allow large tunneling probabilities at the top of barriers. What happens in the double barrier potential problem is *resonant tunneling*: the tunneling probability through the double barriers approaches unity when the incident particle energy aligns with the eigen-energies that exist within the barriers. Essentially, at resonance, the double barriers become transparent.

The same general approach is taken to solve this problem as the previous two with the addition of extra regions. The region descriptions and potential are located in Figure 1.3. Regions I, III, and V have general solutions from Case I and Regions II and IV have general solutions from Case II. Since the bulk of the work is algebraic and the same as the previous problems, it is left out for brevity. The transmission coefficient was found to be:

$$|T|^2 = \frac{1}{1 + \frac{(k^2 + \kappa^2)^2}{4(k\kappa)^4} \sinh^2 \kappa a [2k\kappa \cosh \kappa a \cos kb - (k^2 - \kappa^2) \sinh \kappa a \sin kb]} \quad (1.23)$$

Quantized resonances are then found by setting the equation to unity and solving for the energies.

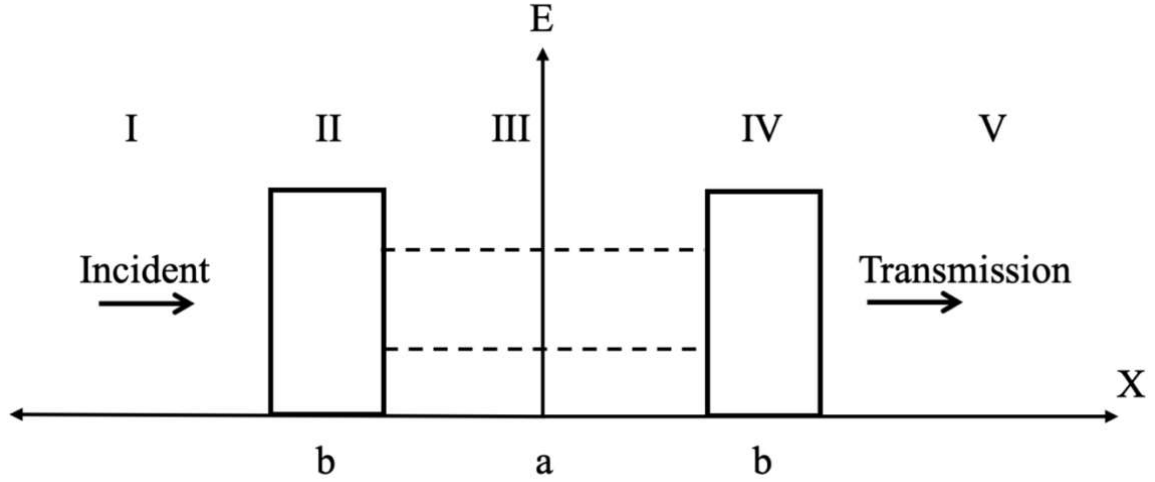


Figure 1.3 A double finite-barrier potential of barrier width ‘b’, well width ‘a’ and resonant energies

1.3: Double Barrier Resonant Tunneling Diode

The double-barrier RTD is an electronic device first simulated by Tsu, Esaki and Chang^[32] that utilizes resonant tunneling discussed in section 1.2.3 for its operation and the defining characteristic of *negative differential resistance*. Before discussing the electrical operation of an RTD a few definitions will be given and shown visually in Figure 1.4.

Negative differential resistance (NDR): A phenomenon by which the increase in applied external potential produces a decrease in absolute current.

Peak-to-valley current-ratio (PVCR): A quality metric for RTDs that is calculated by dividing the peak current (I_p) by the valley current (I_v). The peak and valley currents are defined as the current values at the onset and end of the NDR region.

Voltage and current span: The absolute value difference between the voltage at peak current and valley current and the absolute value difference between the peak current

and valley current. These two metrics are important for the output power metric discussed in the introduction.

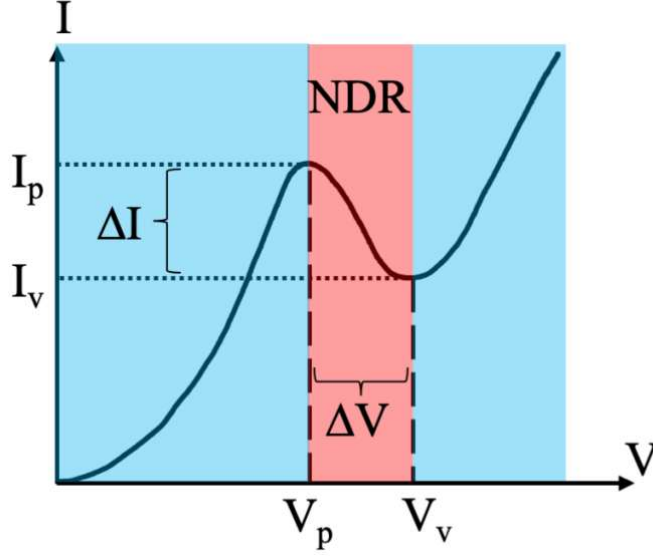


Figure 1.4 A typical current-voltage relationship showing negative differential resistance

The manifestation of NDR in a double barrier RTD is initially evident from the ideal case discussion in section 1.2.3. A quick Gedanken-experiment can show this. Imagine a constant stream of incident particles, a current, to the double barrier potential that slowly moves up in energy. The current would increase to a maximum at one of the resonant energy levels and decrease as the incident beam energy is increased; transmission probabilities are unity only at discrete incident energy levels and are below unity, but non zero, elsewhere. Analytically, the current density depends on the wavefunction and this Gedanken-experiment holds true theoretically. The current density equation in one dimension is:

$$J = \frac{\hbar}{2mi} \left[\psi^* \frac{d\psi}{dx} - \frac{d\psi^*}{dx} \psi \right] \quad (1.24).$$

The difference with a physical RTD is band bending induced by external potentials to set in motion charge carriers. An operational flow for a physical RTD is displayed in Figure 1.5. First [Figure 1.5(a)], the device is in equilibrium and the emitter energy is below the first resonance eigen-energy. Second [Figure 1.5(b)], the applied bias raises the emitter bands and energy level into alignment with the first eigen-energy and the increased current, by way of increased transmission probability, is visualized in the I-V curve. Third [Figure 1.5(c)], the absolute current falls with continual increase in applied potential due to the misalignment of the emitter energy with the eigen-energy. This is the manifestation of NDR in a physical RTD. Lastly [Figure 1.5(d)], the external potential is increased such that the bands have bent enough to lower the second barrier below the emitter energy and current begins to flow by thermal diffusion.

When going from an ideal quantum mechanical problem to a practical device, some additional factors must be discussed. Charge carriers in semiconductor devices obey Fermi-Dirac statistics and density of states given by the materials energy band structure. The result is a distribution of carriers in energy or momentum that can align with the resonant eigen-energies. Furthermore, the charge carriers can scatter off atoms in the lattice that further spread the incident particle momentum and add to excess currents that contribute to the valley current.

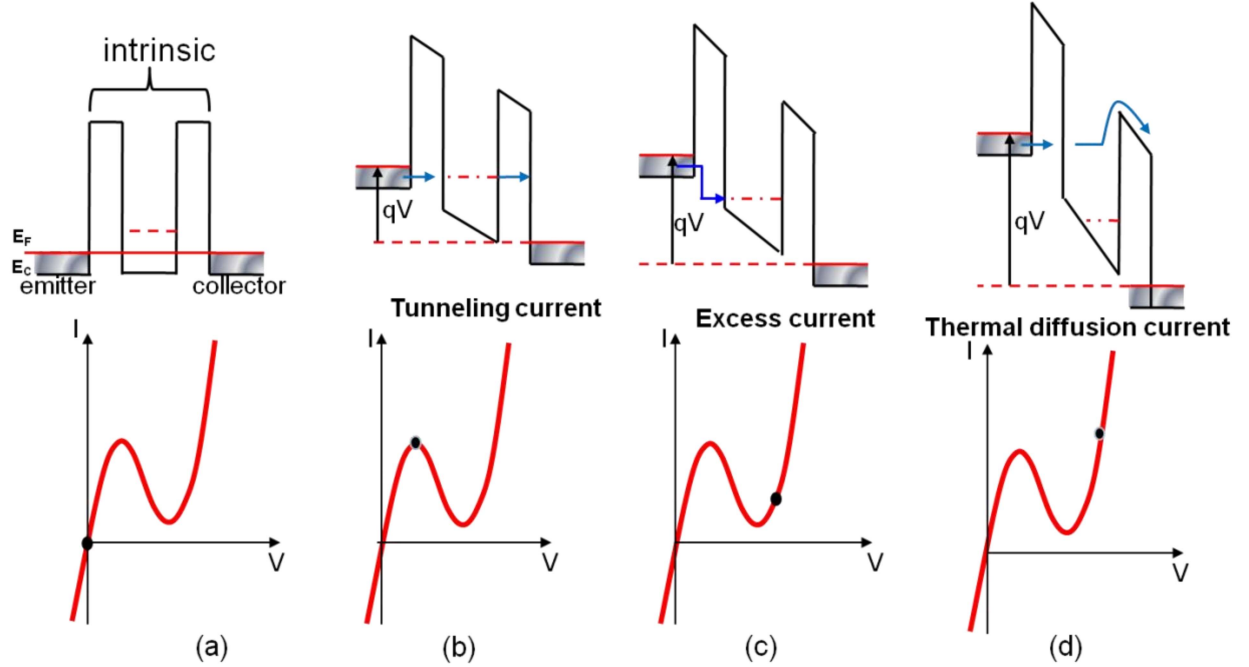


Figure 1.5 Operational flow of a double barrier resonant tunneling diode. (a) Equilibrium structure, (b) Device bias aligns the emitter level with the first confined energy level, (c) Scattering assisted tunneling when the emitter is above the confined energy state, (d) Thermal diffusion regime. Adapted from *Negative Differential Resistance Devices and Circuits*, P R Berger and A Ramesh (2011), adapted from [29]

1.4: Physical Resonant Tunneling Diodes

An RTD can be produced by growing a heterostructure using two materials that have a conduction band, or valence band, offset. The structure discussed in the following chapter uses GaN for the contact layers and well region and AlN for the barrier material. An energy-band diagram for the structure is shown in Figure 1.6. GaN and AlN are piezoelectric materials that induce polarization charges. These charges induce electric fields that manifest into a non-symmetrical, non-rectangular double barrier potential that further deviates the device from ideal.

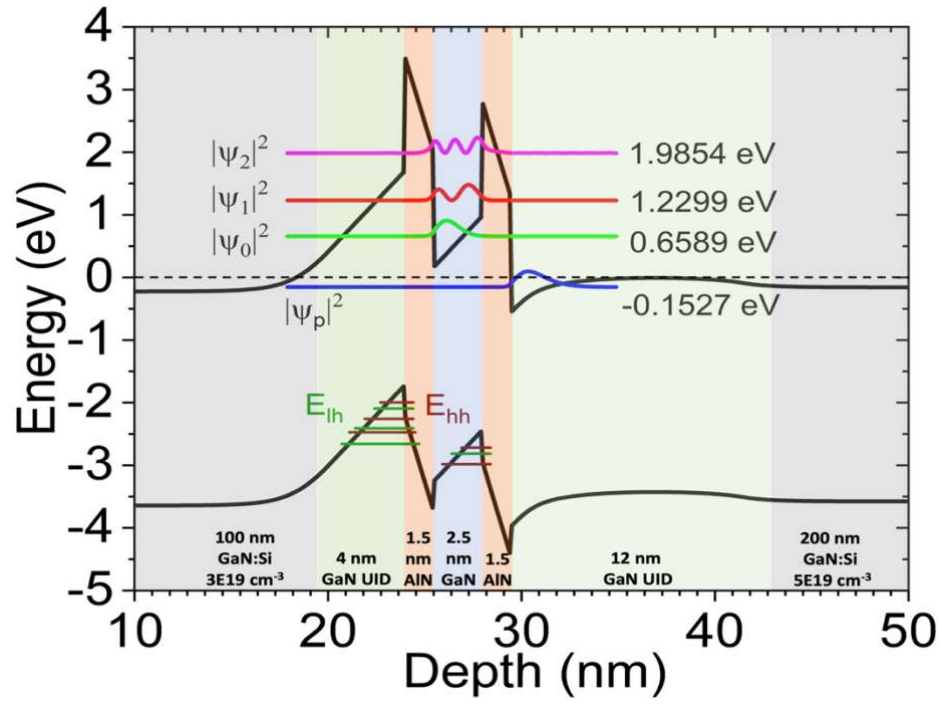


Figure 1.6 Energy band diagram for a typical GaN/AlN double barrier RTD that shows the non-symmetrical nature of a real device structure.

2. GaN/AlN Resonant Tunneling Diode Experiment

2.1: Growth Experiment

The double-barrier RTD structure illustrated in Figure 2.1(a) was grown by plasma-assisted molecular beam epitaxy (PAMBE) in an Omicron/Scienta PRO-75 MBE system equipped with a Veeco® Uni-Bulb™ rf-plasma source for active nitrogen, dual-filament effusion cells for evaporation of elemental Ga and Al, a medium-high temperature effusion cell for Si dopant source, and *in situ* reflection-high-energy electron diffraction (RHEED). The substrate temperature was monitored by a thermocouple positioned behind the molybdenum wafer mount. All samples were grown on $18 \times 18 \text{ mm}^2$ squares diced from a single 50 mm-diameter freestanding GaN wafer obtained commercially. The freestanding, Fe-doped GaN wafer was semi-insulating, with a nominal resistivity in excess of $1 \text{ M}\Omega\text{-cm}$, threading dislocation density, $\delta \sim 3 \times 10^6 \text{ cm}^{-2}$, and thickness of $350 \text{ }\mu\text{m} \pm 25 \text{ }\mu\text{m}$. The Ga-polar (0001) *c*-plane, offcut $0.35^\circ \pm 0.15^\circ$ toward the *m*-plane, was used for growth. The surfaces were prepared with an aggressive *ex situ* HF, HCl, and solvent-based wet chemical clean described in detail elsewhere ^[24,25] immediately prior to loading. Each substrate was degassed under ultra-high vacuum (UHV) conditions for 30 min at 600°C before being transferred to the deposition chamber.

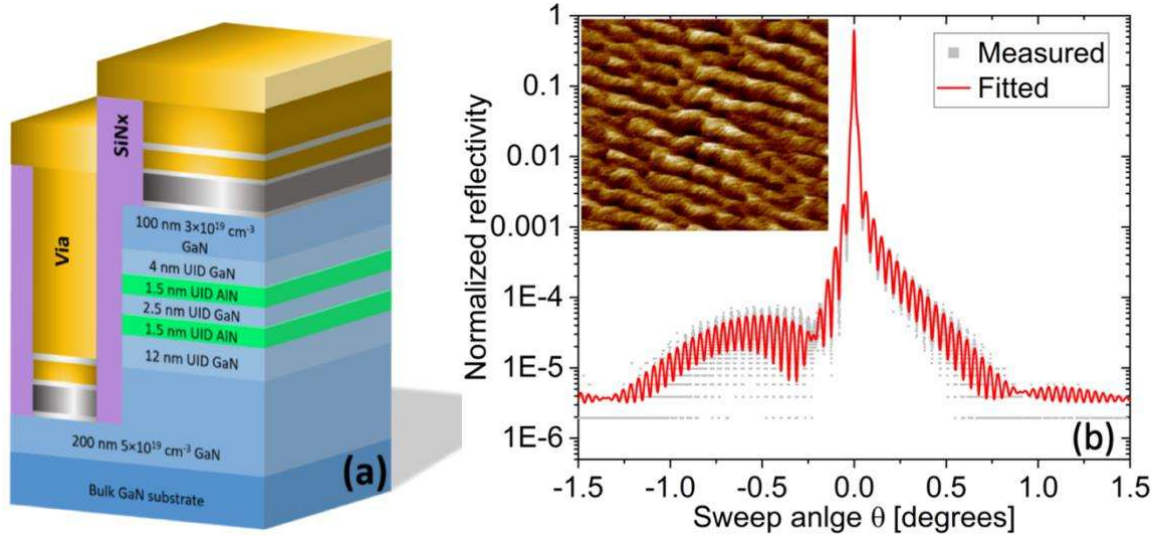


Figure 2.1 (a) Device layer stack. (b) XRD and (inset) $5 \times 5 \mu\text{m}^2$ AFM scan region for Sample B with rms surface roughness of 0.340nm

Once transferred to the deposition chamber, the substrate was ramped to the growth temperature over 20 min. Four samples were grown at substrate temperatures of 760 °C, 810 °C, 860 °C, and 900 °C, respectively. 2×1 RHEED patterns were observed from the GaN substrate surfaces at 760 °C, 810 °C, and 860 °C, but not at 900 °C, where only a streaky 1×1 pattern was observed. The samples are referred as Sample A, B, C, and D, respectively. Growth was initiated by first exposing the substrate surface to the nitrogen plasma for 2 min, during which the 2×1 reconstructed RHEED pattern faded, and the Ga and Si shutters were then opened. The nitrogen plasma was operated at a constant power and flow of 275 W and 0.80 sccm, respectively, resulting in N-limited growth rate of ~ 3 nm/min. The ratio of the Ga and active N fluxes was estimated to be $\sim 1.3:1$. Each sample was grown continuously, without any interrupt, including the four GaN/AlN heterointerfaces.

Table 2.1 The table indicates the sample ID, substrate temperature measured by thermocouple and rms surface roughness by AFM. Also displayed are XRD best-fit parameters and TEM image analysis values for AlN barriers and the GaN quantum well region.

<i>Sample ID</i>	<i>T_s (°C)</i>	<i>AFM: R_q (nm)</i>	<i>FWHM GaN (0002) (arc-sec)</i>	<i>XRD t_{AlN} (nm)</i>	<i>XRD t_{GaN} (nm)</i>	<i>TEM t_{AlN} (nm)</i>	<i>TEM t_{GaN} (nm)</i>
A	760	0.58 ± 0.08	40.3	1.64	2.73	1.64	2.68
B	810	0.37 ± 0.01	28.1	1.55	2.55	1.71	2.22
C	860	3.1 ± 0.9	30.6	1.60	2.68	1.55	2.45
D	900	3.6 ± 0.4	32.0	1.59	2.64	N/A	N/A

Sample A exhibited Ga droplets across the entire sample; the fraction of the surface area covered by droplets was estimated to be 4%. Droplets were observed in a small, isolated region of Sample B, and not at all on Samples C or D. We have separately observed rapid Ga desorption from GaN surfaces at substrate temperatures near 900 °C. [26] The as-grown samples were characterized by atomic force microscopy (AFM) and x-ray diffraction (XRD). The Ga droplets were removed by etching for 3-5 min in HCl. Portions of each sample were cleaved and removed prior to device fabrication for observation by transmission electron microscopy (TEM).

XRD over 3° (ω -2 θ) centered on the GaN [0002] peak was performed on each sample and subsequently fitted using the dynamical simulation software “MadMax”. [28] XRD experimental data and fit for Sample B are shown in Figure 2.1(b). The Pendellösung fringes visible here were present on each sample and allowed for accurate extraction of the GaN growth rate: 3.06-3.14 nm/min. Quantum well and barrier thicknesses were determined from the dynamical simulations of the XRD data, as shown in Table 2.1, given the assumption that AlN grows at the same rate as GaN. Results for barrier and well regions

should not be taken as exact, but they do provide confidence that the layers were grown within the target thickness range (± 1 monolayer).

Surface morphology for each sample was measured by atomic force microscopy (AFM), with the inlaid image in Figure 2.1(b) being from Sample B. Measurements were taken over $5 \times 5 \mu\text{m}^2$ fields of view at three locations along a diagonal for each sample; rms roughness values are presented in Table 2.1. The smoothest surfaces were observed for Samples A and B grown at 760 °C and 810 °C, respectively; higher growth temperatures resulted in rougher surfaces. Large-angle bright-field scanning TEM was performed on Samples A, B, and C, as shown in Figures 2.2(a,b,c). Image analysis to accurately determine the barrier and quantum well thicknesses for each sample was performed using ImageJ^[27] software. The results are also shown in Table 2.1. The XRD and TEM data for barrier and quantum well thicknesses were within one monolayer (~ 0.25 nm) of each other. The variations in the barrier thicknesses extracted from the TEM images correlates with the current densities measured for each growth temperature, as illustrated later in Figure 2.3(c).

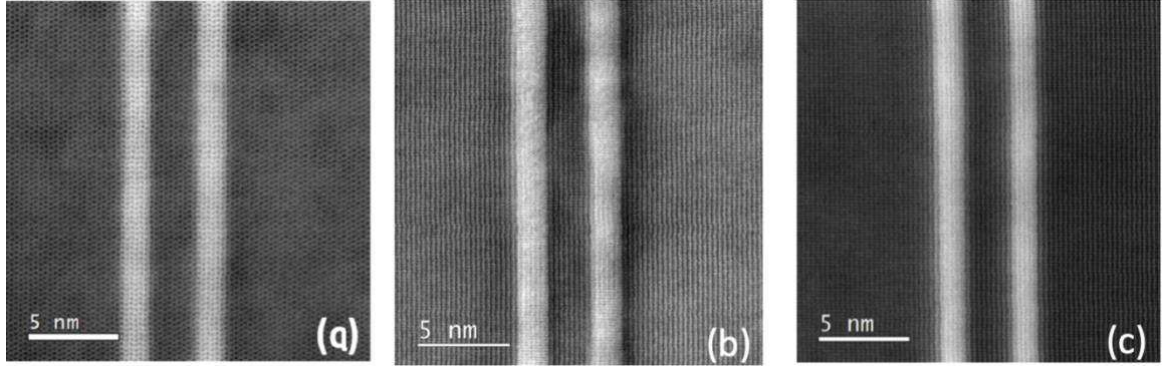


Figure 2.2 Large-angle bright-field scanning TEM images for (a) Sample A, (b) Sample B, and (c) Sample C

Samples A, B and C were co-fabricated into devices, and Sample D was completed later in a lab with different equipment, which should not have affected measured device performance since the fabrication steps for all samples were equivalent. Device fabrication was performed using standard optical lithography and metal liftoff techniques. Mesa definition was performed with a $\text{Cl}_2/\text{BCl}_3/\text{Ar}$ inductively-coupled plasma reactive-ion etch (ICP RIE) producing device areas between $12 \mu\text{m}^2$ and $96 \mu\text{m}^2$ for Samples A, B, and C and between $1.5 \mu\text{m}^2$ and $48 \mu\text{m}^2$ for Sample D. Ti/Al/Ti/Au [25/100/30/50 nm] ohmic contacts were first deposited and annealed using a two-step rapid thermal annealing process. The samples were first annealed at 400°C for 180 seconds and then at 700°C for 18 seconds, both in N_2 ambient, to improve ohmic contact resistance. The circular transmission line method was used for measuring contact resistivity and contact resistance with mean values $5.8 \times 10^{-6} \Omega\text{-cm}^2$ and 0.217Ω , respectively. Device side-wall passivation was then implemented with 250 nm SiN_x deposited by plasma-enhanced chemical vapor deposition. Ti/Au ground-signal-ground (GSG) contact pads were deposited by e-beam evaporation through vias in the passivation layer. A detailed description of the device fabrication can be found in previous reports. ^[13,22]

2.2: Electrical Measurements and Device Analysis

Representative current-voltage (I-V) curves for each size device across all samples are shown in Figure 2.3. The dc I-V measurements were performed by sweeping the applied voltage from -4 V to around +6V depending on location of the NDR region. All four samples showed room temperature NDR. Sample D showed room temperature NDR on devices up to $20\ \mu\text{m}^2$, but then, the differential resistance starting near the expected peak voltage increased and was no longer negative (referred to later as “inflection points”). Samples A and B exhibited room temperature NDR on devices of all sizes between 12 and $96\ \mu\text{m}^2$, albeit with much lower current densities. Samples C and D showed considerably larger currents than Samples A and B. Inflections in the I-V separate from the main NDR region are present for Samples A and B between 2 V and 3 V but are absent from Samples C and D, as shown in Figures 2.3(a-d). These inflection points suggest resonant tunneling through lower quantum-well energy levels with smaller current density. Another possibility is that there are multiple injection mechanisms, as has been previously reported.^[22] Samples C and D failed to show this behavior, likely because the current is dominated by current mechanisms other than resonant tunneling.

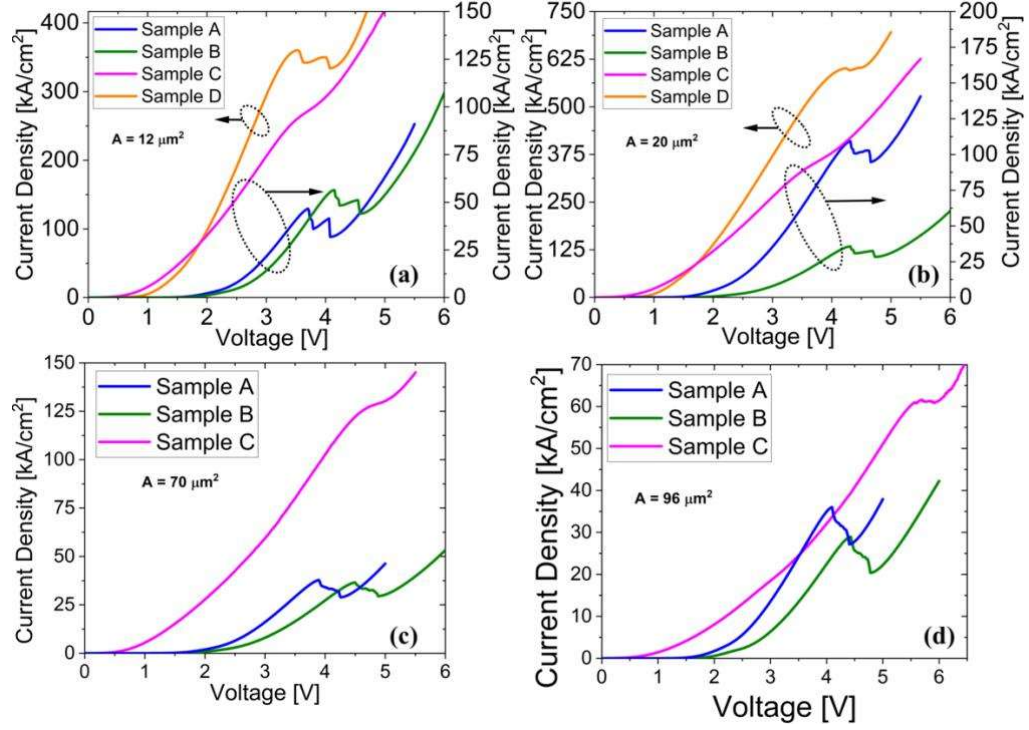


Figure 2.3 Typical current-voltage characteristic for devices of a certain size for increasing growth temperature. (a) $12 \mu\text{m}^2$ devices, (b) $20 \mu\text{m}^2$ devices, (c) $70 \mu\text{m}^2$ devices on all samples excluding Sample D, (d) $96 \mu\text{m}^2$ devices on all samples excluding Sample D.

Several figures-of-merit were used to characterize the quality of the RTD, namely: peak current density (J_p), yield, and peak-to-valley current ratio (PVCR) and peak voltage (V_p). The first three of these are displayed in Figure 2.4 as a function of substrate growth temperature. The averages and standard deviations for J_p and the PVCR were calculated by measuring all working devices on each sample, where a working device was defined as having room-temperature NDR. Increased growth temperature came at some cost, namely, the yield of working devices decreased rapidly from a maximum of $\sim 80\%$ for Sample B down to $\sim 2\%$ for Samples C and D, as shown in Figure 2.4 (b). Average PVCR values followed the same trend as yield, peaking at a value of ~ 1.26 for 810°C (Sample B), as

illustrated in Figure 2.4 (a). An interesting feature of these figures-of-merit was the identical trend shared between PVCR and yield. In general, a large yield is the result of two factors: (1) high growth and fabrication *quality*, and (2) *uniformity* of material across the sample. In contrast, the PVCR is usually an indication of design, growth, and fabrication *quality* but does not require material *uniformity* across the sample. That PVCR and yield follow the same trend is attributed to the fact that they both depend primarily on *quality*; namely the growth. The physical mechanisms that determine quality are interface roughness, dislocations, or the presence of any scattering centers that will broaden the electron wavefunction creating non-resonant-tunneling electron transport.

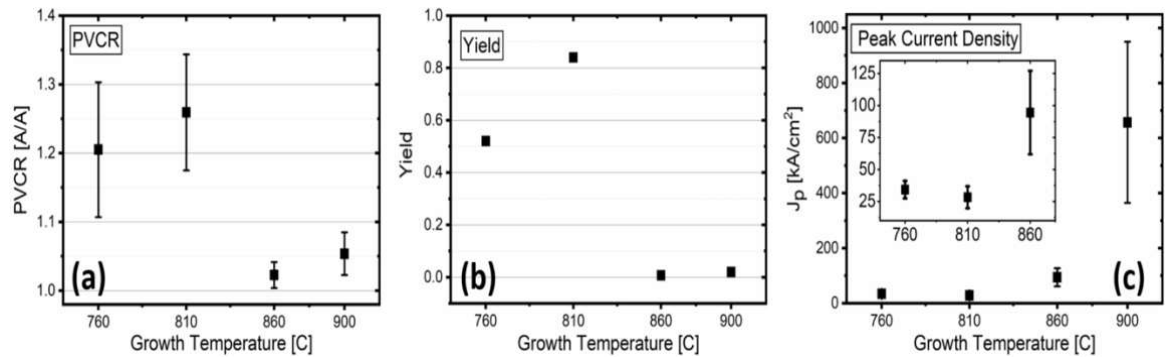


Figure 2.4 Resonant tunneling diode figures-of-merit as function of substrate growth temperature: (a) peak-to-valley current ratio, (b) yield of working devices, (c) peak current density

When the growth temperature was increased from 760 °C to 900 °C, the J_p increased by an order of magnitude, consistent with the fact that the resonant-tunneling component in an RTD is extremely sensitive to barrier thickness, and also to well thickness to a lesser extent. For the thin barriers (nominally 1.5 nm) used here, reduction in thickness by one monolayer can increase the current density by roughly an order of magnitude.^[20] While the large-angle bright-field scanning TEM images illustrated in Figures 2.2(a-c) are supportive

of this hypothesis, these represent only a snapshot of a small part of the entire device. To verify this would require an exhaustive TEM study that examined a considerable portion of wafers capable of operational devices.

2.3: Effects of device area on device performance

The large number of working devices fabricated for Samples A and B, namely 231 and 293, respectively, allows for statistical investigation. Each figure-of-merit as a function of growth temperature can be plotted as a function of device area as illustrated in Figures 2.5 and 2.6, respectively. Generally, the variance for each distribution describing Sample A is larger than the equivalent distribution in Sample B because of the higher yield for sample B. Variance reduction is proportional to the sample size. Device performance is affected by area in multiple facets. PVCR, peak voltage, and current density all trend toward lower device performance with an increase in device area, as evident in Figures 2.5 and 2.6. The PVCR decreased with area due to self-heating effects: larger absolute current increases device temperature via Joule heating, subsequently increasing carrier scattering events that contribute to the valley current. ^[19] Further confirmation stems from the comparison between PVCR for these two samples and their current densities. The average current density for each size device of Sample A is larger than the equivalent size device of Sample B [Figures 2.5(c),2.6(c)], while having an inverse relationship for PVCR [Figures 2.5(a),2.6(a)].

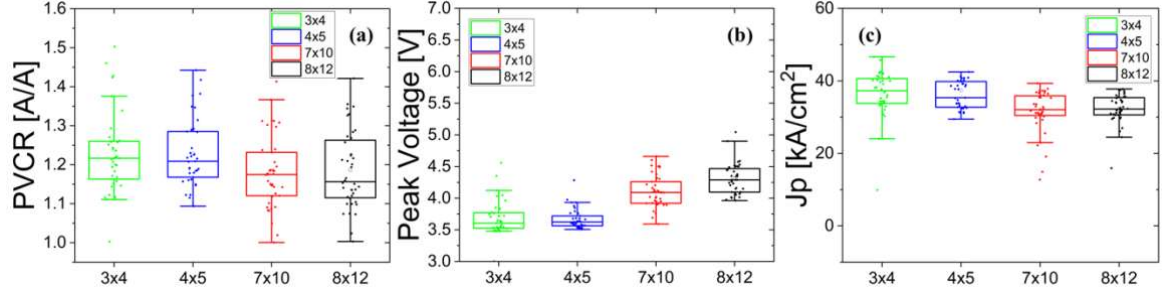


Figure 2.5 Figures-of-merit for sample A versus device area (μm^2). (a) PVCR, (b) peak voltage, and (c) current density at peak voltage

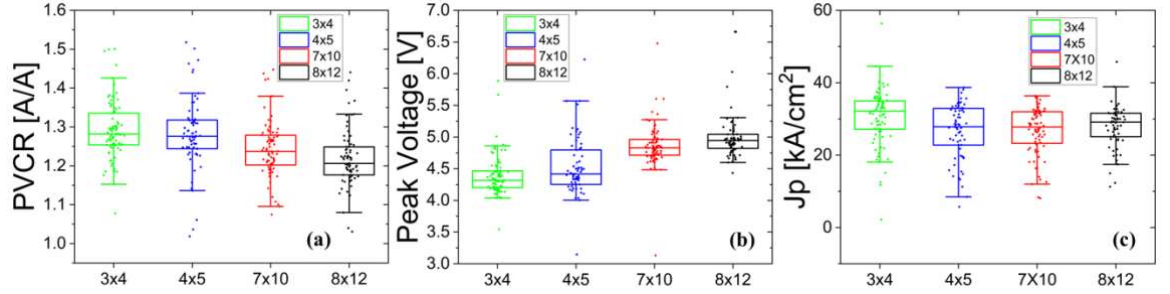


Figure 2.6 Figures-of-merit for sample B versus device area (μm^2). (a) PVCR, (b) peak voltage, and (c) current density at peak voltage

Peak voltage increased with device area and was a result of fixed external series resistance ($\sim 1.5\Omega$) in the test network affecting the measured extrinsic voltage when absolute current increased. For the two smallest area devices, the peak voltage distribution is skewed towards larger values and appears to reach a minimum value $\sim 3.5\text{V}$. Although there are slight general trends in these metrics, it is important to note the relative uniformity of the devices. Accompanied by a large number of working devices, it is reasonable to conclude that these devices are of high quality and repeatable. The device space spanned by growth temperature and device size allow for a tuned device based on the specific application. For example: An RTD-based logic circuit requires large PVCR. A small area device ($< 12 \mu\text{m}^2$) grown close to 810°C is suggested to achieve this requirement.

2.4: Current density record

The previous highest peak current density of 930 kA/cm^2 was measured on nominally the same device structure, changing only the substrate to free-standing bulk GaN.^[19] Reported here is the first GaN-based RTD exhibiting greater than 1 MA/cm^2 peak current density. Sample D was the source of the device. A I-V dc measurement of the device displayed in Figure 2.7 shows the lack of room-temperature NDR. It is suspected that the valley current grows via LO phonon-assisted tunneling due to self-heating from resistive effects as the device area increases and eventually masks the NDR.^[19] Sample D was affected more so than the other samples due to larger absolute current and current density. To reduce the thermal energy increase under steady bias, pulsed measurements were recorded using an Accent DiVA pulsed I-V meter with pulse widths of 200 ns and a duty cycle of $4 \times 10^{-3} \%$, which is in contrast to the previous report of 930 kA/cm^2 measured at 100% duty cycle. Raw pulsed data from the meter was smoothed using a Savitzky-Golay filter, with the result overlaid in Figure 2.7 showing room-temperature NDR. Devices with the largest peak current density on each of the other samples are compared with each other and overlaid in Figure 2.7. Sample D produced other devices with comparable current density. One such device is shown in Figure 2.7 (black curve) displaying a 100%-duty dc J-V peak current density of 976 kA/cm^2 .

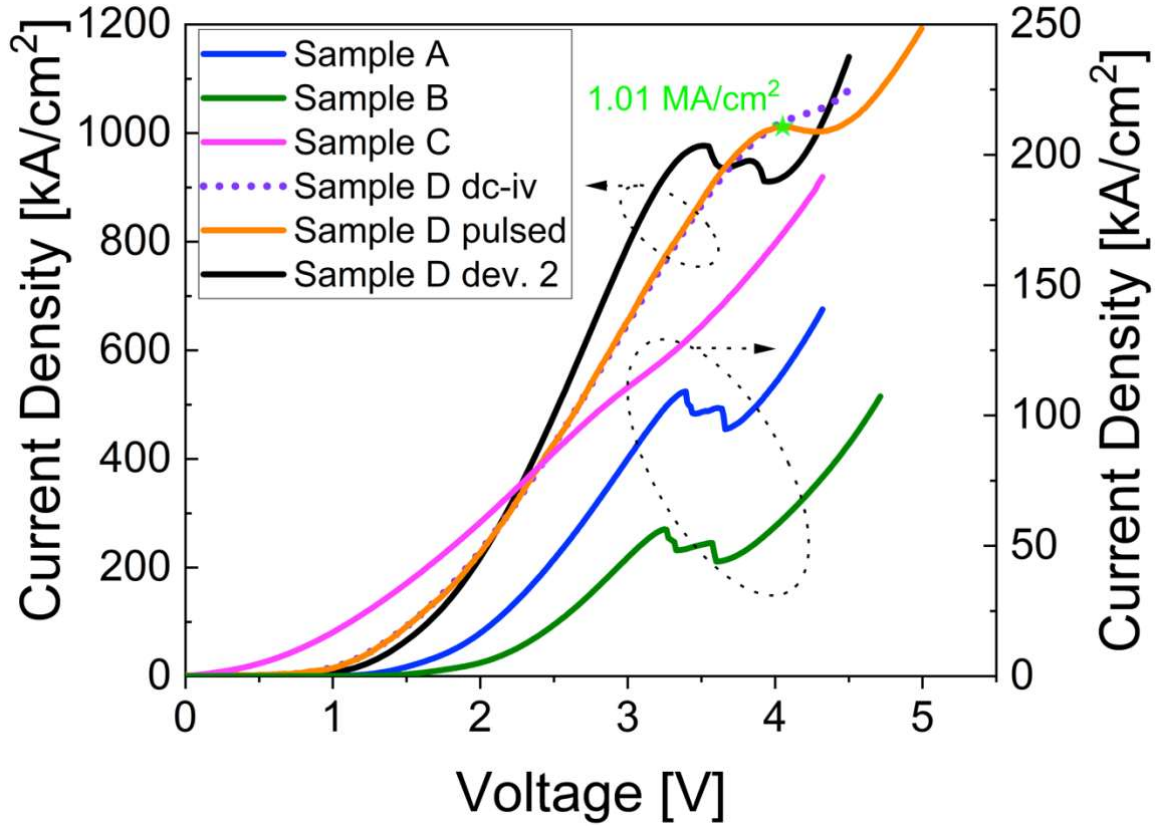


Figure 2.7 J-V curves for devices with largest current density on their respective sample. Included in this figure is the record-breaking sample pulsed I-V measurement and dc I-V measurement.

2.5: Future Plans

The results given in this chapter show how with the growth techniques and conditions, GaN/AlN RTDs have reached a new level of maturity and give vectors for future improvements. The first being pointed towards fabrication techniques. The fabrication methods used in the production of these devices have not been updated in multiple device cycles and contain a lot of room for improvement. A few examples will be given here. Sidewall current leakage is a lingering issue that could be mitigated by improved mesa etching methods, sidewall passivation and device shape. To improve the sidewalls, hotter ICP-RIE etch conditions will be used and wet chemical etching to repair

sidewall damage will be explored. Passivation of the sidewalls will be improved by experimenting with atomic-layer deposition of highly ordered metal-oxides, such as TiO_2 or HfO_2 . Contact photolithography by a Karl-Suss MJB3 was used in the fabrication of these devices and should be replaced as soon as possible. Uniformity of devices is difficult to achieve, and the mask device shape of a rectangle sometimes develops in the photoresist as an abstract shape with wiggles and scalloped edges. E-beam lithography or a mask-less laser-writer is suggested. The benefit of E-beam lithography is, at order zero, two-fold as device size can be reduced to state-of-the-art and will improve the current density as discussed above in Section 2.3.

A second improvement vector is towards epitaxial structure. The current density of the devices studied increased with decrease in barrier thickness and yield was maximum at a growth temperature of 810 °C. A natural progression is a reduction of nominally grown barrier widths while targeting a maximum yield growth temperature to combine the two benefits.

3. Conclusions

An introduction into the quantum mechanical phenomena of particle tunneling through a potential barrier and discrete bound states in finite energy potential wells was given. To understand the operation of a double barrier RTD, first two energy potential problems were solved using Schrödinger's time independent equation. The first two problems described separately particle tunneling through energy barriers by means of a transmission probability and the existence of quantized bound states in a finite-energy potential well by means of quasi-eigenenergies within the well. Once the two phenomena were understood, the themes of the two problems were combined to investigate the double barrier potential. The transmission probability for particles through a double barrier structure was solved for and revealed a probability of unity for incident energies that align with the quasi-eigenenergies that exist between the barriers. This phenomenon is known as *resonant tunneling* and it is of different character than single barrier tunneling and the particles, in the ideal case, do not become bound between the barriers. A Gedanken-experiment followed to show at first-order, how resonant tunneling through a double barrier potential could be utilized in a semiconductor device to produce NDR. Performance metrics of PVCR, peak current, valley current, voltage span and current span were defined for RTDs and the operating cycle was discussed. Moving from the ideal analog to an RTD to a physical device required the discussion of semiconductor charge carriers following Fermi-Dirac statistics, band-structure defined density of states and excess currents from phonon scattering, impurity scattering, and trap assisted tunneling. RTD devices are produced by the growth of heterostructures with large conduction band offset between the

two materials and requires a high quality of crystallinity. Following the theoretical and background discussion was the GaN/AlN RTD growth temperature experiment.

Trends in figures-of-merit for GaN/AlN RTDs as a function of growth temperature and device area have been observed, leading to record high current densities of $>1\text{MA}/\text{cm}^2$. This substrate temperature growth study allowed for statistical analysis of PVCR, yield and peak current density, which indicated that current density generally increased with growth temperature at a cost of decreased yield and PVCR. Samples A and B exhibited the highest yield, which allowed for statistical analysis as a function of device area for PVCR, peak voltage and peak current density. Generally, an increase in device area reduced the device quality in all four metrics. These findings produce a 2-dimensional “phase-space” between growth temperature and device area that allow for application-specific RTD production and add to the viability for high power applications that require large current density. Furthermore, the GaN/AlN RTDs presented here provide a stable foundation for the future work on device fabrication, growth techniques and epitaxial structure.

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